

Amazon S3 Bucket ACL

To Begin with the Lab

Summary of the Lab

In this lab, **S3 Bucket ACLs** were demonstrated as a legacy access-control method for S3 buckets and objects. The video showed that ACLs are disabled by default on new buckets, and that they can be enabled later along with an **Object Ownership** setting such as **Bucket owner enforced** or **Object writer**. It then explained the four main ACL permissions—**List**, **Write**, **Read ACL**, and **Write ACL**—and showed how these permissions apply either to the bucket itself or to the objects inside it. The lab also highlighted that ACLs can grant access to specific AWS accounts or predefined groups, but they are less flexible and harder to manage than bucket policies or IAM policies.

Understanding S3 Bucket ACL

An **ACL (Access Control List)** is an older S3 permission model that gives basic access control at both the bucket level and the object level. It exists so you can grant read or write access to a specific AWS account or to certain predefined groups, but it does not give fine-grained control like modern JSON-based policies do. In this lab, ACLs were shown as useful mainly for legacy scenarios, object ownership handling, and simple permission sharing. The benefit is basic access control, but the downside is that ACLs are harder to manage at scale and easier to misconfigure, which is why AWS recommends IAM policies and bucket policies instead.

Example:

If another AWS account uploads an object into your bucket, the ACL behavior depends on the **Object Ownership** setting. With **Bucket owner enforced**, the bucket owner keeps full control of the uploaded object. With **Object writer**, the uploading account keeps control over its own object permissions.

Lab Steps

Step 1 – Review the default ACL state

- Open **Amazon S3**.
- Create a new bucket.
- Confirm that ACL is **disabled by default** on new buckets.

Amazon S3 > Buckets > Create bucket

Bucket name [Info](#)

cloudfreeks-acl

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn more](#)

Copy settings from existing bucket - optional

Only the bucket settings in the following configuration are copied.

[Choose bucket](#)

Format: s3://bucket/prefix

Object Ownership [Info](#)

Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

Object Ownership

☒ **ACLs disabled (recommended)** [Info](#)
Objects in this bucket can be owned by this account. Access to this bucket and its objects is specified using only policies.

☐ **ACLs enabled**
Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

Object Ownership
Bucket owner enforced

Block Public Access settings for this bucket

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

☒ **Block all public access**
Turning this setting on is the same as turning on all four settings below. Each of the following settings is independent of one another.

☐ Block public access to buckets and objects granted through new access control lists (ACLs)

- Open the bucket **Permissions** tab.

Amazon S3 > Buckets > cloudfreeks-acl

cloudfreeks-acl [Info](#)

[Objects](#) [Metadata](#) [Properties](#) **[Permissions](#)** [Metrics](#) [Management](#) [File systems - new](#) [Access Points](#)

Permissions overview

Access finding
Access findings are provided by IAM external access analyzers. Learn more about [How IAM analyzer findings work](#).
[View analyzer for us-east-1](#)

Block public access (bucket settings) [Edit](#)

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to all your S3 buckets and objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to your buckets or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

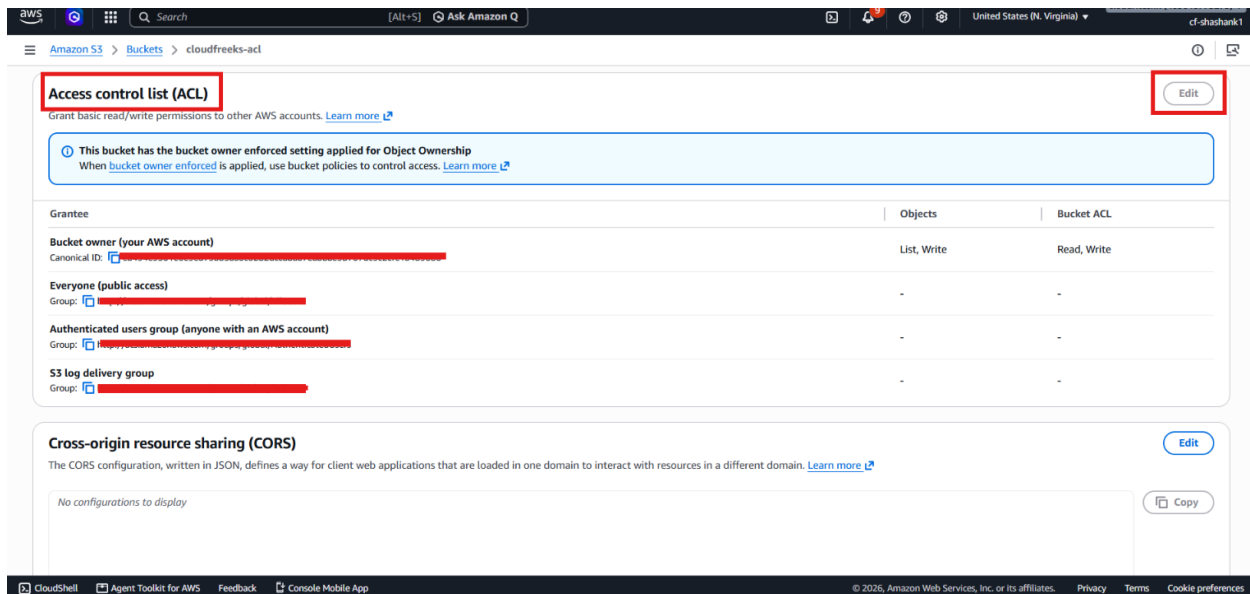
Block all public access
☒ On
► **Individual Block Public Access settings for this bucket**

Bucket policy [Edit](#) [Delete](#)

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. [Learn more](#)

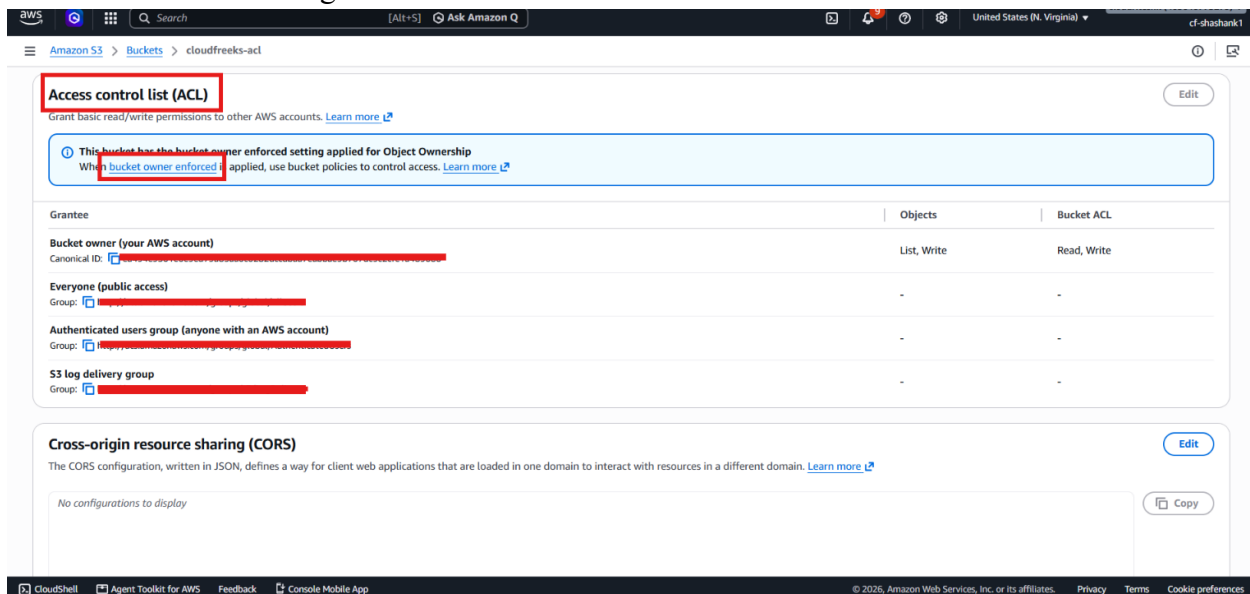
Public access is blocked because Block Public Access settings are turned on for this bucket
To determine which settings are turned on, check your Block Public Access settings for this bucket. Learn more about [using Amazon S3 Block Public Access](#).

- Verify that **Access control list** is greyed out until ACL is enabled.

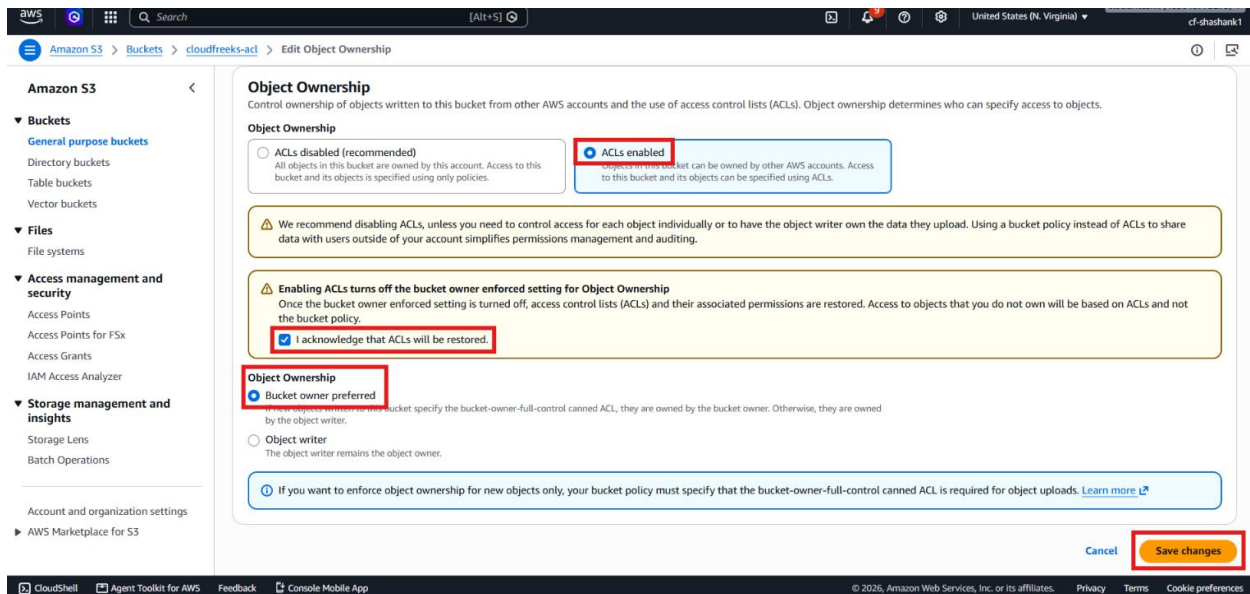


Step 2 – Enable ACL and choose object ownership

- Go to the ACL settings and click on **bucket owner enforced**.

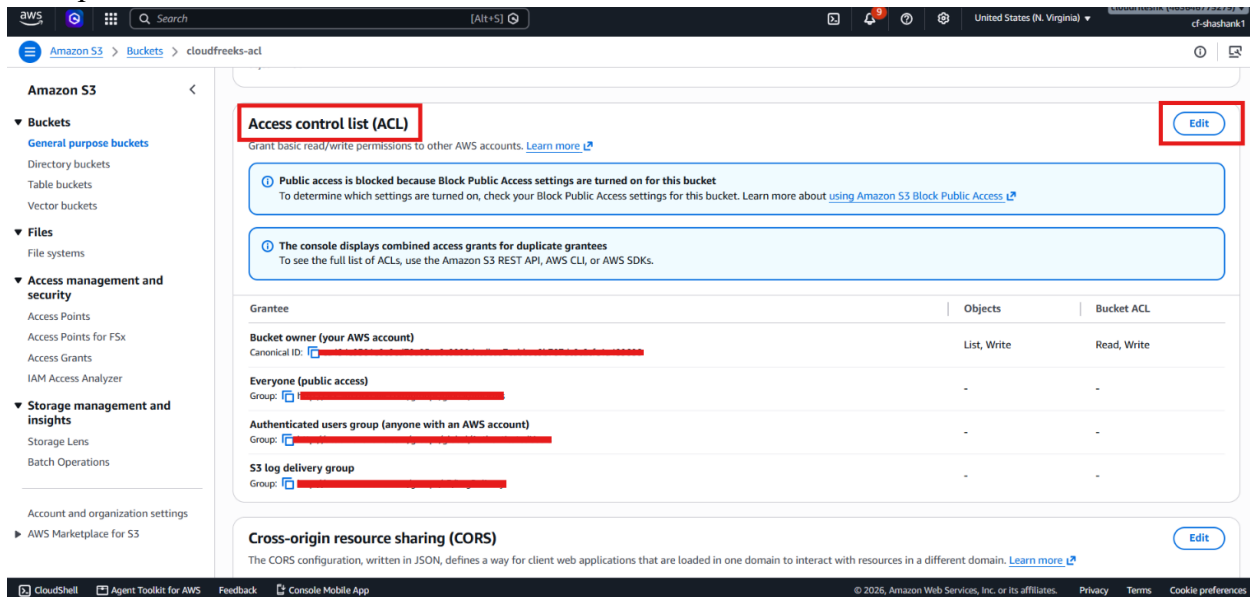


- Select **ACL Enabled** option and acknowledge it.
- Choose **Bucket owner enforced** to keep bucket-owner control over uploaded objects, or **Object writer** to let the uploader keep object control.
- Save the changes.

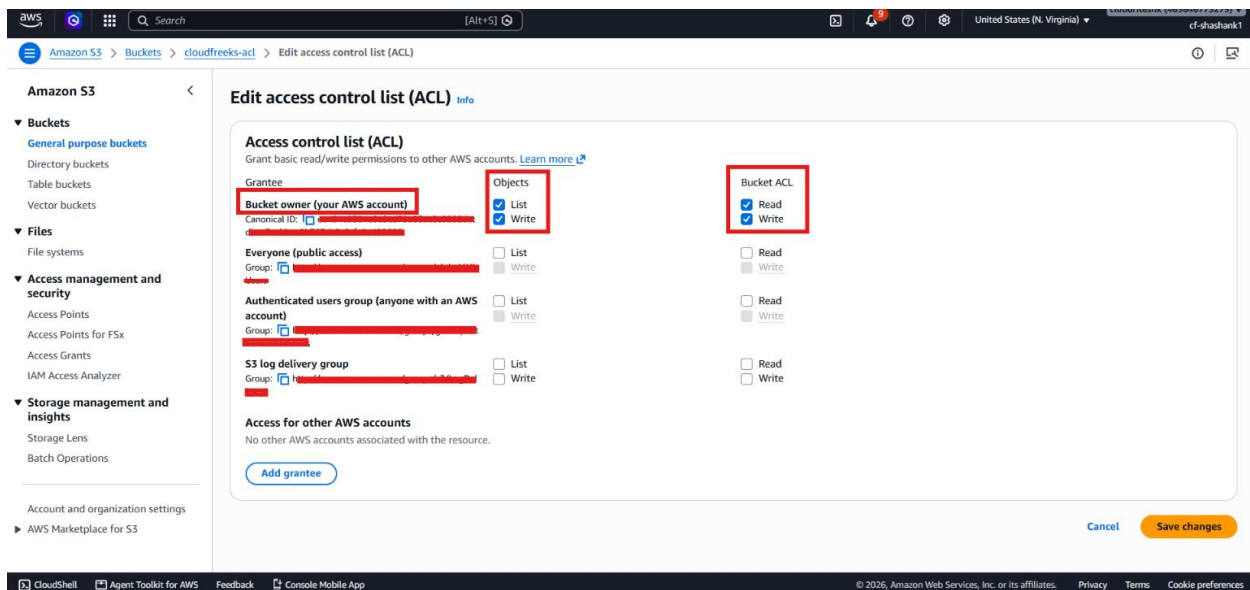


Step 3 – Understand ACL permissions

- Open the bucket **Access control list** section and click on **Edit**.



- Review the four main permissions:
 - **List**
 - **Write**
 - **Read ACL**
 - **Write ACL**
- Understand that **List** shows bucket contents, **Write** allows create/overwrite/delete, **Read ACL** lets you inspect ACL settings, and **Write ACL** lets you modify ACL permissions.



Step 4 – Understand ACL grantees

- Review the predefined ACL groups:
 - Everyone**
 - Authenticated users**
 - S3 log delivery group**
- Understand that ACL can also grant access to a specific AWS account by using the account's **canonical user ID**.
- Note that ACL does not provide permission control for IAM users, groups, or roles directly.

Step 5 – Review the limitations

- Recognize that ACLs offer only basic permissions.
- Understand that they are difficult to manage at scale.
- Note that they are easy to misconfigure.
- Remember that ACLs are an older access-control method compared with IAM and bucket policies.

Verification

- Verified that ACLs are disabled by default on new S3 buckets.
- Verified that ACLs can be enabled later from the bucket permissions.
- Verified that **Bucket owner enforced** and **Object writer** control who owns uploaded objects.
- Verified that ACL permissions include **List**, **Write**, **Read ACL**, and **Write ACL**.
- Verified that ACLs can grant access to specific AWS accounts or predefined groups.
- Verified that ACLs are a legacy and less scalable method compared with IAM policies and bucket policies.